

# ZAKI ILIAS

Data Engineer | Microsoft Fabric & Azure | Apache Airflow | dbt | Power BI

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## PROFESSIONAL SUMMARY

Data Engineer specialized in building multi-source pipelines on Microsoft Fabric, with 1 year of production data engineering experience within a cybersecurity environment (SOC). Full command of the data value chain: API ingestion (REST, GraphQL), Delta Lake modeling, Fabric pipeline orchestration, and Power BI reporting. Proven autonomy in international environments. Available for a permanent Data Engineer position from September 2026.

## WORK EXPERIENCE

**Data Engineer & Data Analyst** CTDI Europe | Strasbourg, France | Work-Study Contract (hybrid) 09/2025 – Present

- Optimized SOC data model: redesigned architecture, improved performance and reliability of cybersecurity data pipelines
- Integrated 4 new cybersecurity data sources (identity protection, pentesting, SIEM alerts, endpoint alerts) via fully automated Python API pipelines
- Delivered production architecture: 6 data sources, 18 Delta Lake tables, 5 orchestrated Microsoft Fabric pipelines
- Designed Power BI reports for the SOC team: security KPIs, real-time alerts, dynamic filters

**Data Engineer** CTDI GmbH | Malsch, Germany | On-site Internship (4 months) 05/2025 – 08/2025

- Built from scratch a production-ready multi-source SOC data pipeline, integrating and consolidating data from multiple cybersecurity APIs
- Delivered to production: 7 Delta Lake tables in Azure Fabric Lakehouse, fully automated archiving pipelines

**Intern — Civil Engineering Site Supervision** SBTX | Drriouch Province, Morocco | Internship (2 months) 07/2022 – 08/2022

Civil engineering execution studies and on-site construction supervision — first professional experience.

## EDUCATION

**Master of Science, Big Data (In Progress)** JUNIA ISEN | Lille, France 2025 – 2026

**Master of Science, Big Data** JUNIA ISEN | Lille, France 2024 – 2025

**1st Year Engineering Cycle — Top of Class** ISEN | Rabat, Morocco 2023 – 2024

**Preparatory Cycle — Top of Class** ISEN | Rabat, Morocco 2021 – 2023

## SKILLS

**Data Engineering & BI:** Python, pandas, REST APIs, GraphQL, Delta Lake, SQL, dbt, star schema, Microsoft Fabric, Power BI, Azure Data Lake, Terraform

**Orchestration & Big Data:** Apache Airflow, Apache Spark (PySpark), Hadoop, Docker, Docker Compose, Azure Container Apps

**Machine Learning & AI:** scikit-learn, XGBoost, CNN (MobileNetV2), Vision Transformer (ViT), Transfer Learning, R, Jupyter

**Development & DevOps:** FastAPI, Django, PHP, MySQL, GitHub Actions, CI/CD, OAuth2, OpenCV, EasyOCR

**Languages:** French (native bilingual), English (bilingual — TOEIC 850/990), Spanish (beginner)

## CERTIFICATIONS

**Microsoft (Coursera):** Harnessing Data with Power BI • ETL in Power BI • Data Modeling in Power BI • Preparing Data for Analysis with Power BI

**Stanford University / DeepLearning.AI (Coursera):** Supervised Machine Learning: Regression and Classification

**Imperial College London (Coursera):** Mathematics for Machine Learning: Linear Algebra

**University of Michigan (Coursera):** Programming for Everybody (Getting Started with Python) • Python Data Structures

## PROJECTS — GITHUB.COM/ILIASZAKI

**Agora** | Python, Apache Airflow, dbt, PostgreSQL — End-to-end DE pipeline: synthetic e-commerce data (1,500–3,000 orders/day), Airflow orchestration, dbt transformations, star schema

**Cloud Transcriber** | Azure Container Apps, Terraform, Docker, FastAPI, Whisper, GitHub Actions — Audio transcription app on Azure, infrastructure as code, full CI/CD pipeline

**Weather App** | FastAPI, Docker Compose, Nginx, OpenWeatherMap, GitHub Actions — Full-stack 3-service weather app, automated cache, GitHub Container Registry CI/CD

**ML Padel Project** | Python, scikit-learn, XGBoost, FFT, MFCC, Hilbert — Custom signal processing pipeline (FFT, MFCC, Hilbert) on dual-modality sensor data (audio + vibration); 4 models × 4 feature sets × 3 classification objectives (impact zone, racket type, age); up to 100 % accuracy

**Blockchain from Scratch** | Python, RSA, SHA-256, P2P — Full blockchain: PoW & PoS, RSA wallets, on-chain smart contracts, NFTs, multi-node P2P network

**QuickQCM** | PHP, MySQL, Python, OpenCV, EasyOCR — QCM platform with automated OCR-based grading and Excel report generation